

Final Closeout Report

Closeout Report - MPC as a Layer 2 Service to Cardano

Project ID: 1200045 | Fund: Project Catalyst

Proposer: Profila AG

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1. Executive Summary

This Proof of Concept demonstrates that Cardano smart contracts (Aiken) can coordinate with Partisia Blockchain's MPC infrastructure to run privacy-preserving analytics on personal data. The PoC implements a complete cross-chain architecture: an Aiken validator on Cardano Preprod records the MPC computation request, an off-chain relay bridges the request to Partisia Blockchain, a ZK MPC contract on PBC testnet receives secret user data and computes an aggregate result, and the result is displayed without exposing any individual data at any layer.

2. Milestones

ID	Title	Status	Key Evidence
M1	Aiken Smart Contract on Cardano	Complete	cardano/validators/profila_mpc.ak
M2	Partisia ZK Contract + Off-Chain Relay	Complete	partisia/contracts/profila_mpc/src/
M3	Secret Input Flow & Privacy Verification	Complete	test-data/profila_test_users.json
M4	MPC Computation + Result Display + E2E + Complete	Complete	relay/show_result.ts

M1 - Aiken Smart Contract on Cardano [Complete]

Deliverables:

- Aiken MpcRequest validator with InitiateMPC redeemer
- 6 unit tests (all passing via aiken check)
- Deployed to Cardano Preprod with inline datum
- TX hash: 50d28fd2bd263a84485b45280afd19bb4c3e20c24e9a4b52b6eac20418d53a57

Evidence:

- cardano/validators/profila_mpc.ak
- cardano/scripts/deploy-m1.json
- docs/evidence/milestone-1-report.json

M2 - Partisia ZK Contract + Off-Chain Relay [Complete]

Deliverables:

- PBC ZK contract (contract.rs + zk_compute.rs) with age_threshold and survey_match computations
- 11 unit tests (all passing via cargo test)
- Off-chain relay (relay.ts) polling Blockfrost and bridging to PBC
- Contract deployed to PBC testnet via browser

Evidence:

- partisia/contracts/profila_mpc/src/
- relay/relay.ts

- relay/relay-log.json
- partisia/deploy/deploy-m3.json

M3 - Secret Input Flow & Privacy Verification [Complete]

Deliverables:

- Synthetic dataset (50 users, 35 over age 18, 28 survey=true)
- Real secret submission via PBC TypeScript SDK - 5 confirmed on-chain ZK transactions
- Privacy verification: no individual data in contract state, secret tab, logs, or relay output

Evidence:

- test-data/profila_test_users.json
- relay/submit_secrets.ts
- relay/secret-submissions-log.json
- docs/evidence/milestone-3-report.json

M4 - MPC Computation + Result Display + E2E + Closeout [Complete]

Deliverables:

- show_result.ts fetches live PBC state via REST API, decodes ProfilaMpcState from on-chain binary
- trigger_computation.ts documents computation trigger (browser-based due to CLI limitation)
- 5 real secret-input transactions submitted to PBC contract via @partisiablockchain/zk-client SDK
- result-display.html with cross-chain evidence and live on-chain state
- E2E run log documenting all 6 stages with real TX hashes
- Closeout report (this document)

Evidence:

- relay/show_result.ts
 - relay/submit_secrets.ts
 - relay/trigger_computation.ts
 - relay/mpc-result.json
 - relay/secret-submissions-log.json
 - relay/result-display.html
 - docs/evidence/e2e-run-log.json
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3. Architecture

Layer 1

Chain: Cardano Preprod

Contract Type: Aiken validator

Role: On-chain anchor for MPC computation request

Script Address: addr_test1wzut662xhd8e4jq97fpdsml2pyqerejwzfg24red2n7gceczhkwaz

Bridge

Component: Off-chain relay (TypeScript)

Role: Polls Blockfrost for new UTXOs, parses datum, bridges to PBC

Pattern: Standard PBC second-layer bridging pattern

Layer 2

Chain: Partisia Blockchain Testnet

Contract Type: ZK MPC contract (Rust)

Role: Receives secret inputs via MPC secret sharing, computes aggregate result

Contract Address: 039ed9214602f2a93eb05411f852fa78a476607634

4. Key Achievements

- Demonstrated Cardano-to-PBC cross-chain coordination via off-chain relay
 - Deployed a ZK MPC contract on PBC testnet with verifiable on-chain state
 - Submitted 5 real ZK secret-input transactions via @partisiablockchain/zk-client TypeScript SDK
 - Successfully triggered ZK MPC computation across 4 nodes - on-chain result: 5 (all submitted users over age 18)
 - Proved that secret user data never appears in any on-chain state, log, or output
 - Built binary state decoder that reads ProfilaMpcState directly from PBC REST API
 - Created comprehensive evidence trail with real TX hashes on both chains
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5. Technical Notes

Pbc Cli Limitation

The cargo pbc transaction CLI returns HTTP 400 for all transaction types on PBC testnet. Contract deployment uses the PBC testnet browser. Secret input submission and computation trigger use the PBC TypeScript SDK.

Pbc Typescript Sdk

Secret inputs submitted via @partisiablockchain/zk-client (RealZkClient.buildOnChainInputTransaction) + @partisiablockchain/blockchain-api-transaction-client (BlockchainTransactionClient.signAndSend). Computation trigger uses SDK with corrected RPC encoding (0x09 prefix byte for action invocations). Each secret transaction encrypts the value for all 4 MPC engines.

Zk Gas Requirement

ZK MPC computation across 4 nodes requires ~100M gas. The computation trigger transaction must include sufficient gas fee; excess goes to the contract's balance.

Pbc Sdk Version

v.16.126.0 (binder 11.9.0) - upgraded from v15.1.0 for testnet binder compatibility

Pbc Rest Api

Contract state accessible via GET /chain/contracts/{address} - the /blockchain/contracts/ endpoint returns 404

Binary State Decoding

ProfilaMpcState decoded by locating the administrator address anchor in the serializedContract base64 blob, then reading fields sequentially per PBC's ReadWriteState format

Secret Submissions

5 real on-chain ZK secret-input transactions submitted via PBC TypeScript SDK. Each transaction creates encrypted shares for the 4 MPC engines. Transaction hashes verifiable on PBC testnet browser.

6. Test Summary

Suite	Total	Passed	Framework
Aiken (Cardano)	6	6	aiken check
PBC (Rust)	11	11	cargo test
Total	17	17	All passing

Aiken Tests

- test_initiate_mpc_valid
- test_initiate_mpc_empty_dataset
- test_initiate_mpc_wrong_signer
- test_initiate_mpc_bad_query
- test_initiate_mpc_stale_timestamp
- test_initiate_mpc_no_datum

PBC Tests

- test_init_contract_valid_query_types
 - test_init_contract_rejects_invalid_query_types
 - test_init_contract_min_participants_validation
 - test_compute_age_threshold_result
 - test_compute_survey_match_result
 - test_decode_zero_result
 - test_decode_large_result
 - test_decode_short_byte_slice
 - test_decode_empty_slice
 - test_survey_sum_result_decoding
 - test_query_type_dispatch_logic
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7. Videos

Final Closeout Video (Catalyst PCV):

<https://youtu.be/YFOK4pq9jIk>

M1 - Aiken Contract:

<https://youtu.be/QRj4tAt23Aw>

M2 - PBC Contract + Relay:

https://youtu.be/IC_2EG6gxX0

M3 - Secret Inputs:

<https://youtu.be/psm2IMU7n1o>

M4 - Result & Closeout:

<https://youtu.be/2YYvELDzrGo>

Combined M1-M4 End-to-End:

<https://youtu.be/jwwfdTSUS2w>

8. Repository

URL:

https://github.com/Profila/Aiken-Partisia_MPC

Branch: main

Files: ~30 source files across cardano/, partisia/, relay/, test-data/, docs/

9. Future Work

- Optimize gas estimation for ZK MPC computation trigger (currently requires 100M gas)
 - Add Cardano result callback (write MPC result back to Cardano UTxO)
 - Production deployment with real Profila user data (with consent)
 - Multi-query support (run age_threshold and survey_match in parallel)
 - Mainnet deployment on both Cardano and PBC
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*Profila AG x Partisia Blockchain
Project Catalyst Fund 13 - Proposal #1200045*